

Supporting Engineering Workflows - Flow E, Flow F, Flow G, and SOC Dashboard Rollup

The engineering maturity layer: direct MCP policy monitoring, red-team replay/regression harness, false-positive analytics, rule tuning recommendations, Slack digests, and dashboard metrics for the MVP V2 operating model.

Document Type: Workflow project document | Supporting workflows

Prototype Focus: Policy monitor | Regression | False positives | Dashboard analytics

Design Goal: Show that the prototype is measurable, tunable, and regression-aware beyond one-time alert generation.

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Why this document matters

- The write-up is designed as both a portfolio-grade project document and an implementation-facing prototype note.
- It explains the problem, the workflow logic, the SOC value, and the validation evidence captured from real lab executions.
- The project extends the previous MVP V1 pattern into MCP, RAG/memory, and agentic AI security operations.

Key outcome	Why it matters
Flow E	Direct MCP policy event monitoring before or alongside full Wazuh integration.
Flow F	Replay/regression harness proving detection behavior is repeatable.
Flow G	False-positive analytics and tuning recommendations from case/runtime data.
Dashboard	Daily/weekly V2 metrics rollup across A2/B2/C2/E/F/G evidence streams.

Architecture view

Supporting Engineering Workflows

Quality, tuning, and metrics layer for the MVP V2 operating model



MVP V2 prototype: production-inspired design, validated in lab, prepared for portfolio and further hardening.

Architecture view - Supporting workflows add quality, tuning, and metrics around the core A2/B2/C2 lifecycle.

Project Summary

The supporting workflows are the quality and analytics layer for the MVP V2 prototype. Flow E monitors MCP policy events directly; Flow F reruns the red-team regression corpus; Flow G analyzes false positives and tuning opportunities; the dashboard rollup summarizes the overall SOC posture.

Problem Statement

One successful alert does not prove a detection program is healthy. Teams need policy debugging, repeatable regression, false-positive analysis, noisy-rule tracking, tuning recommendations, and daily/weekly metrics. These supporting flows turn the prototype into a more realistic security operations program.

Solution Overview

These workflows surround A2/B2/C2 with engineering feedback loops. They make V2 measurable: policy violations, replay pass rate, case closure quality, false-positive rates, rules needing tuning, and dashboard metrics become visible.

Tools and Technologies Used

Tool	Role in the prototype
Flow E	Webhook-driven MCP policy monitor for direct app/policy layer events.
Flow F	ExecuteCommand regression harness with Slack report and regression DataTable.
Flow G	TheHive/runtime join logic for FP rate and tuning recommendations.
Dashboard Rollup	Aggregates MCP, RAG/memory, agentic, A2/B2, regression, and false-positive metrics.
Slack	Digest notifications for policy violations, regression results, tuning, and dashboard rollups.
DataTables	Audit rows for policy monitor, regression runs, false-positive analytics, recommendations, and dashboard metrics.

How the System Works

Stage	What happens
Policy monitor	Flow E receives MCP policy events directly and records violations.
Replay harness	Flow F reruns a controlled corpus and compares expected rule IDs and pass/fail outcomes.
FP analytics	Flow G reads closure/runtime data and computes FP rate by rule/family.
Dashboard	Rollup workflow builds metric rows for SOC posture reporting.
Evidence	Slack and DataTables prove each supporting workflow executed successfully.

SOC Value and Expected Impact

- Improves detection reliability by catching regressions before content changes are trusted.
- Reduces alert fatigue by creating a pathway for false-positive rate tracking and tuning recommendations.

- Improves leadership visibility through dashboard metrics, not only raw alert screenshots.
- Supports sustainable SOC operations by adding feedback loops around policy, detection, triage, and closure.

Implementation and Validation Evidence

The supporting workflow evidence shows each analytics/quality workflow running successfully, posting Slack messages, and updating DataTables.

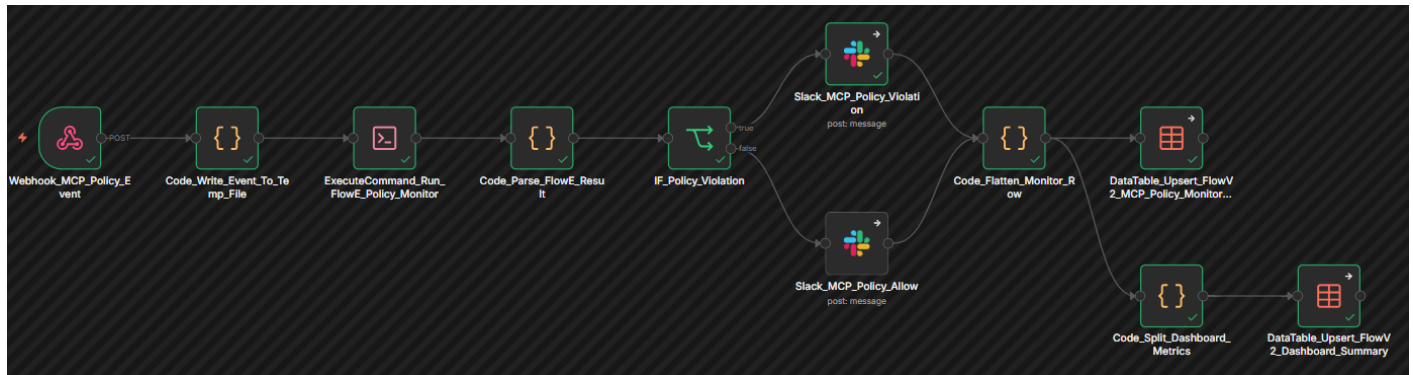


Figure S-01 - Flow E MCP Runtime Policy Monitor workflow canvas.

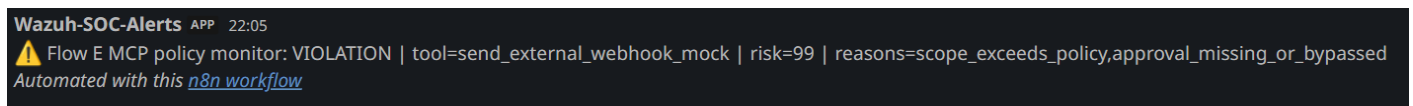


Figure S-02 - Slack policy violation notification from Flow E.

Data tables / flow_v2_mcp_policy_monito...							Q Search	Add Row	Add Column
T monitor_event_id	timestamp	T request_id	T session_id	T user_id	T agent_id	T mcp_server_name			
flowe-3b351e144e4ef521	2026-05-09T05:30:00.000+05:30	evidence-flow-e-policy-violation-001	sess-flow-e-evidence-001	student-user-demo	agent-demo-001	local-security-tools			

Figure S-03 - Flow E DataTable evidence for MCP policy event / violation row.

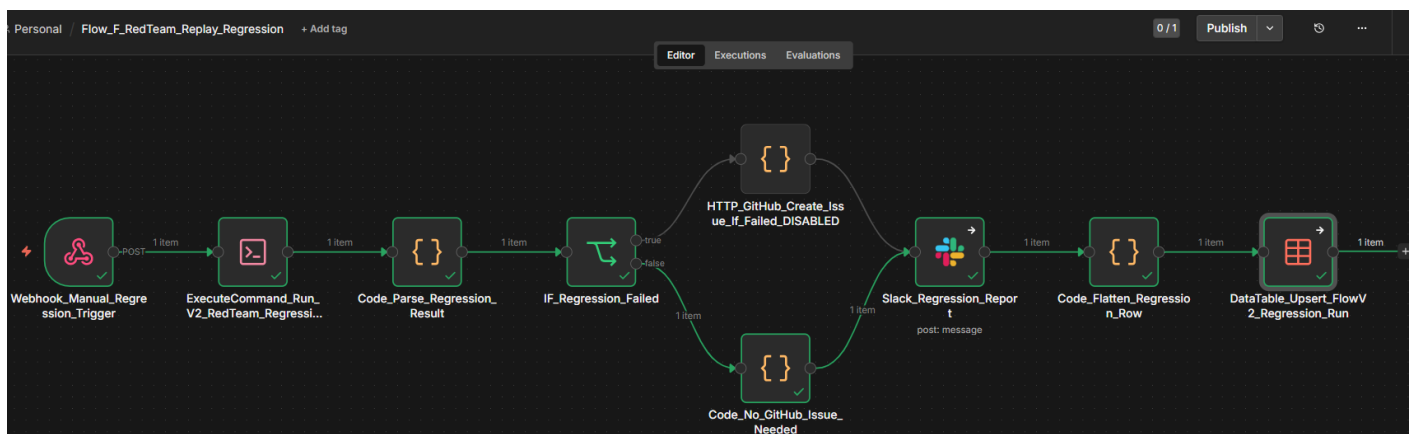


Figure S-04 - Flow F Red-Team Replay and Regression workflow canvas.

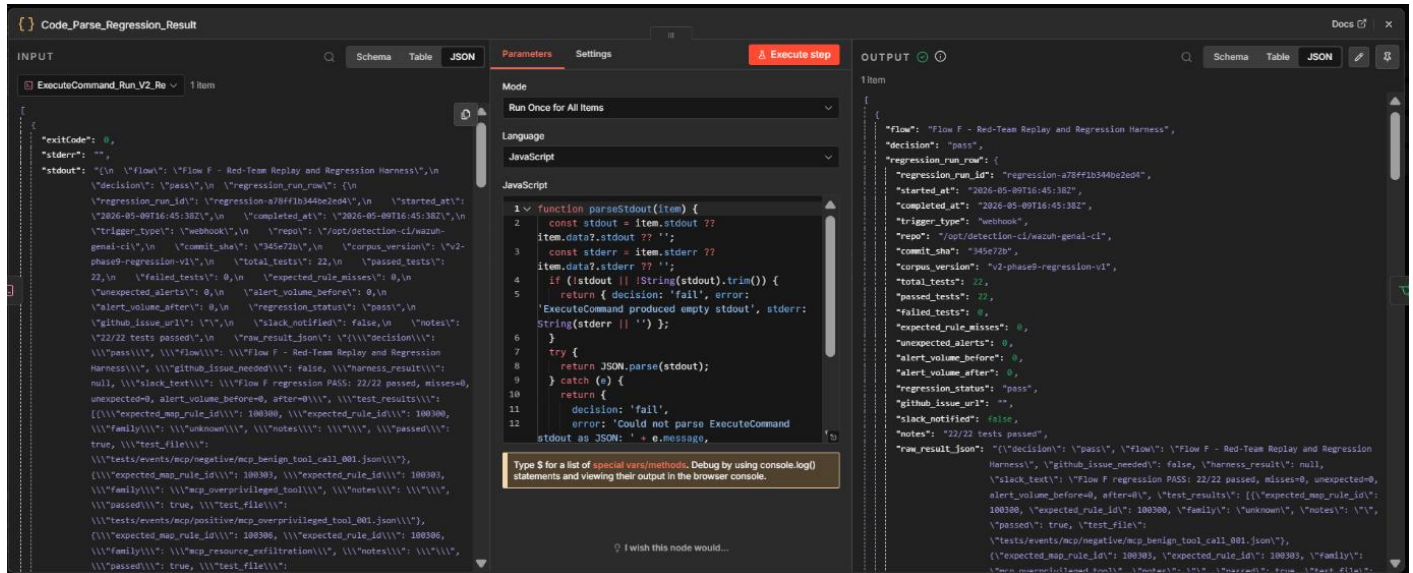


Figure S-05 - Flow F replay harness command/output node evidence.

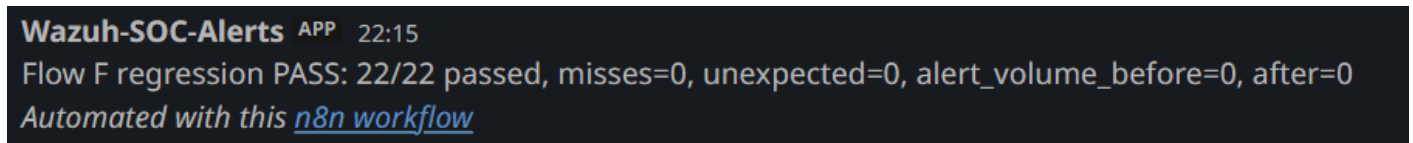


Figure S-06 - Slack regression PASS notification showing 22/22 tests passed.

T regression_run_id	started_at	completed_at	T trigger_type	T repo	T commit_sha	T corpus_version	# total_tests	# passed_tests
regression-0774ee51d19828f	2026-05-09T22:57:30	2026-05-09T22:57:30	webhook	/opt/detection-ci/wazuh-genai-ci	345e72b	v2-phase9-regression-v1	22	22

Figure S-07 - Flow F regression DataTable row after upsert fix.

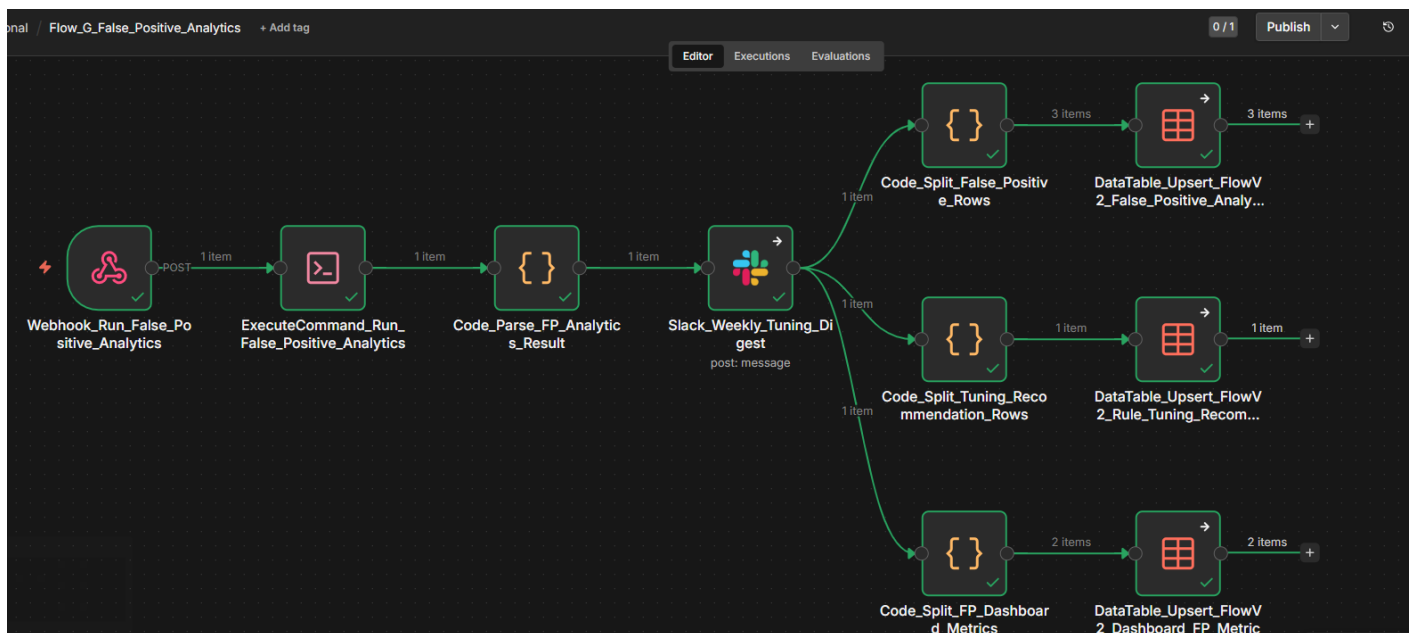


Figure S-08 - Flow G False Positive Analytics workflow canvas.

The screenshot shows the 'Code_Parse_FP_Analytics_Result' node in a workflow editor. The node is configured with Mode 'Run Once for All Items' and Language 'JavaScript'. The output shows a JSON object with various analytics metrics like rule_family, alert_count, case_count, and fp_rate.

Figure S-09 - Flow G analytics calculation node output for false-positive/tuning logic.

Wazuh-SOC-Alerts APP 22:33

Flow G FP analytics: rules=3, recommendations=1. 100301 fp=0.5 action=critical; 100351 fp=0.0 action=low; 100402 fp=0.0 action=low

Automated with this [n8n workflow](#)

Figure S-10 - Slack weekly tuning digest from Flow G.

Data tables / flow_v2_rule_tuning_recom... :

T recommendation_id	created_at	# rule_id	T rule_family	T priority	T reason	T recommended_action	# fp_rate	# alert_count
tune-485a36a8129a8eac	2026-05-09T22:33:28	100 301	mcp_tool_poisoning	critical	fp_rate=0.5; alerts=2	High FP rate: review rule logic, tighten c	0.5	2

Figure S-11 - Flow G false-positive analytics DataTable evidence.

Data tables / flow_v2_false_positive_anal... :

T analytics_id	window_start	window_end	# rule_id	T rule_family	# alert_count	# case_count	# closed_false_positive_count	# closed_true_positive_count	# closed_benign_positive_count	# fp_rate
fp-20790d1b41bfd710	2026-05-02T22:33:28	2026-05-09T22:33:2	100 301	mcp_tool_poisoning	2	2	1	1	0	0.5
fp-b8785bee4d9ba3bc	2026-05-02T22:33:28	2026-05-09T22:33:2	100 351	agent_goal_hijack	1	1	0	1	0	0
fp-0ab2a0d20d01a0d	2026-05-02T22:33:28	2026-05-09T22:33:2	100 402	memory_poisoning_attempt	1	1	0	0	1	0

Figure S-12 - Flow G rule tuning recommendation DataTable evidence.

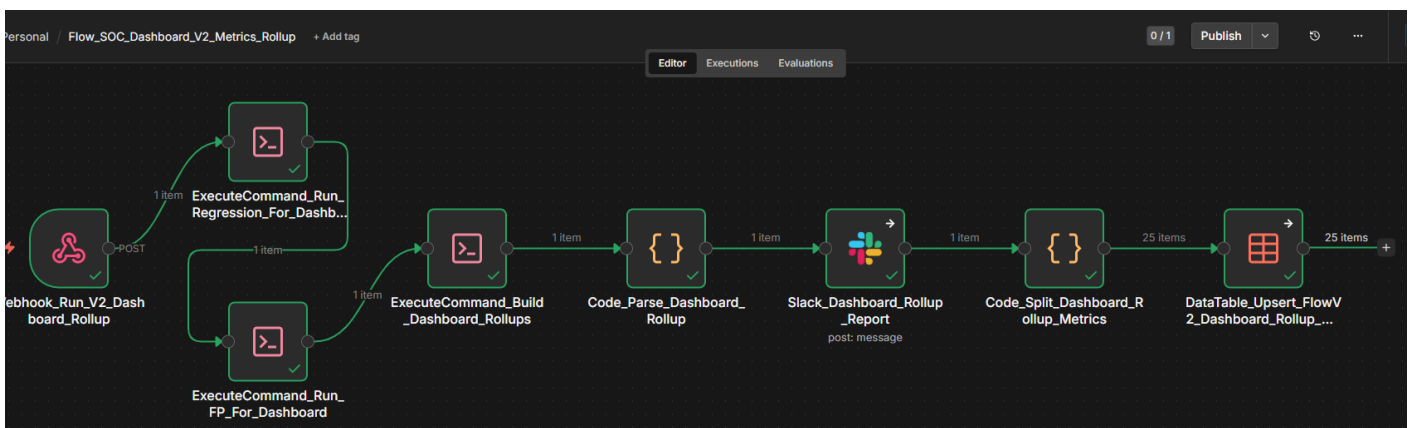


Figure S-13 - SOC Dashboard V2 Metrics Rollup workflow canvas.

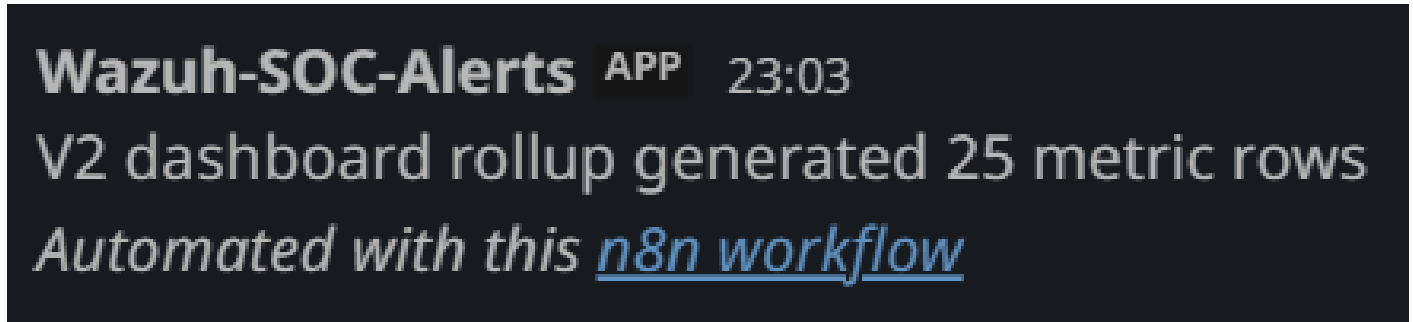


Figure S-14 - Slack dashboard rollup notification showing generated metric rows.

Personal / Data tables / flow_v2_dashboard_summary									
Search Add Row									
T metric_id	window_start	window_end	T metric_name	# metric_value	T dimension_key	T dimension_value	T source_table	generated_at	T notes
metric-8cab0e3eb2e34c7e	2026-05-09T05:30:0	2026-05-09T22:05:56	mcp_policy_events_total	1	flow	flow_e	flow_v2_mcp_policy_monitor_events	2026-05-09T22:05:	Flow E direct policy event
metric-4a010ba806735d43	2026-05-02T22:33:2	2026-05-09T22:33:28	false_positive_rate_by_rule	0.1667	aggregate	all_rules	flow_v2_false_positive_analytics	2026-05-09T22:33:	Average FP rate across rules in window
metric-251960043a46d497	2026-05-02T22:33:2	2026-05-09T22:33:28	rules_needing_tuning	1	priority	medium_or_higher	flow_v2_rule_tuning_recommendations	2026-05-09T22:33:	Rules with tuning recommendations
metric-368679327568733c	2026-05-09T05:30:0	2026-05-09T23:03:33	regression_pass_rate	1	aggregate	all	flow_v2_phase9_regression_runs	2026-05-09T23:03:	22/22 tests passed
metric-0240d7b3748fe421	2026-05-09T05:30:0	2026-05-09T23:03:33	regression_failed_tests	0	aggregate	all	flow_v2_phase9_regression_runs	2026-05-09T23:03:	Empty
metric-27474fa65c2d88ab	2026-05-09T05:30:0	2026-05-09T23:03:33	alert_volume_before	0	aggregate	all	flow_v2_phase9_regression_runs	2026-05-09T23:03:	Empty
metric-e46752ae7e96b9a7	2026-05-09T05:30:0	2026-05-09T23:03:33	alert_volume_after	0	aggregate	all	flow_v2_phase9_regression_runs	2026-05-09T23:03:	Empty
metric-3e3050579ba65dfb	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_tool_calls_total	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-5e93f38680fcc252	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_tool_calls_blocked	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-fd4e9c726da9cf2	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_tool_results_sanitized	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-7ee82b25e323f57d	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_schema_drift_detected	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-878f2c652d6b53d5	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_tool_poisoning_detected	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-e2b02741b9a42054	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_sensitive_tool_without_hilt	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-54d225ce875f239e	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_untrusted_server_detected	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-554e69683a8bc2bf	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_resource_exfiltration_attempts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-83a9f66dd9aa1804	2026-05-09T05:30:0	2026-05-09T23:03:33	mcp_confused_deputy_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-ab64b48f1ecd1bed	2026-05-09T05:30:0	2026-05-09T23:03:33	rag_untrusted_context_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-3e5de60efb7bb076	2026-05-09T05:30:0	2026-05-09T23:03:33	rag_retrieval_to_tool_escalations	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-6149f0fabf3cd259	2026-05-09T05:30:0	2026-05-09T23:03:33	memory_poisoning_attempts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-6d895fb202059ceb	2026-05-09T05:30:0	2026-05-09T23:03:33	memory_scope_violations	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-2f5309fee1591137	2026-05-09T05:30:0	2026-05-09T23:03:33	agentic_goal_hijack_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-51098ca16794c36c	2026-05-09T05:30:0	2026-05-09T23:03:33	agentic_tool_loop_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-edcc4ad724f21192	2026-05-09T05:30:0	2026-05-09T23:03:33	approval_manipulation_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-bc1e491eecc7700d	2026-05-09T05:30:0	2026-05-09T23:03:33	delegated_identity_misuse_alerts	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-03943cc714f2cbc0	2026-05-09T05:30:0	2026-05-09T23:03:33	case_promotions_by_family	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e
metric-f55595f73047d0bc	2026-05-09T05:30:0	2026-05-09T23:03:33	top_noisy_rule_family	0	aggregate	all	phase9_rollup_placeholder	2026-05-09T23:03:	Populate from n8n DataTable reads or e

Figure S-15 - Dashboard rollup DataTable with V2 metric rows.

Limitations and Production Hardening Roadmap

- Flow E should eventually authenticate and sign policy events from real applications.
- Flow F regression should expand from the MVP corpus into larger red-team and benign corpora with continuous scheduled runs.
- Flow G needs production-quality closure taxonomy, analyst disposition discipline, and enough case volume for statistically meaningful FP rates.
- Dashboard rollup should be connected to a visualization layer and alert-level retention policy for production operations.

Skills Demonstrated

Skill area	Evidence in this build
Detection QA	Built replay/regression logic and corrected DataTable schema/upsert behavior.

Metrics engineering	Produced dashboard rows for MCP/RAG/agentive, regression, false-positive, and deployment metrics.
SOC tuning	Designed false-positive analytics and rule-tuning recommendation flow.
Policy operations	Built direct MCP policy monitor as a pre-Wazuh debugging path.

Conclusion

The supporting workflows make MVP V2 feel like a real detection engineering and SOC operations program. They prove the system can be tested, tuned, measured, and improved over time.

Together with A2, B2, and C2, these workflows complete the prototype operating model.

Document prepared for portfolio showcase by [abdul4rehman215](#)